DefinITely, the most boring learning content you will ever see!

Jan Bielecki, PyCon Wrocław 2026

Intro

More Philosophy than Python



Learning platform based on DefinIT project



k4liber.github.io/definit-dsa

Learning platform based on DefinIT project



"Perfection is achieved, not when there is nothing more to add, but when there is nothing left to take away." ~ Antoine de Saint-Exupéry

Demo

Nobody:

Bugs right before a demo:



Showcase of http://k4liber.github.io/definit-dsa

About me

Jan Bielecki (1994 - ?)



Senior Full Stack Developer at Ørsted (2023 - ?)

Jan Bielecki (1994 - ?)



"Any sufficiently advanced technology is indistinguishable from magic." ~ Arthur C. Clarke

Jan Bielecki (1994 - ?)





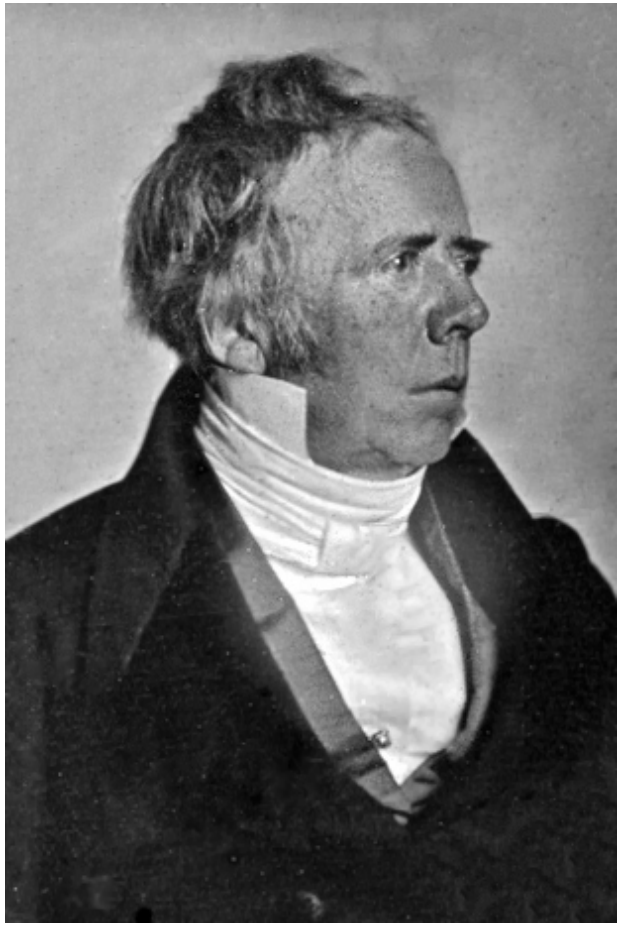




Running, Calistenics, Ping-pong, Foosball, Trekking, Snowboarding

About Ørsted

Hans Christian Ørsted (1777 - 1851)





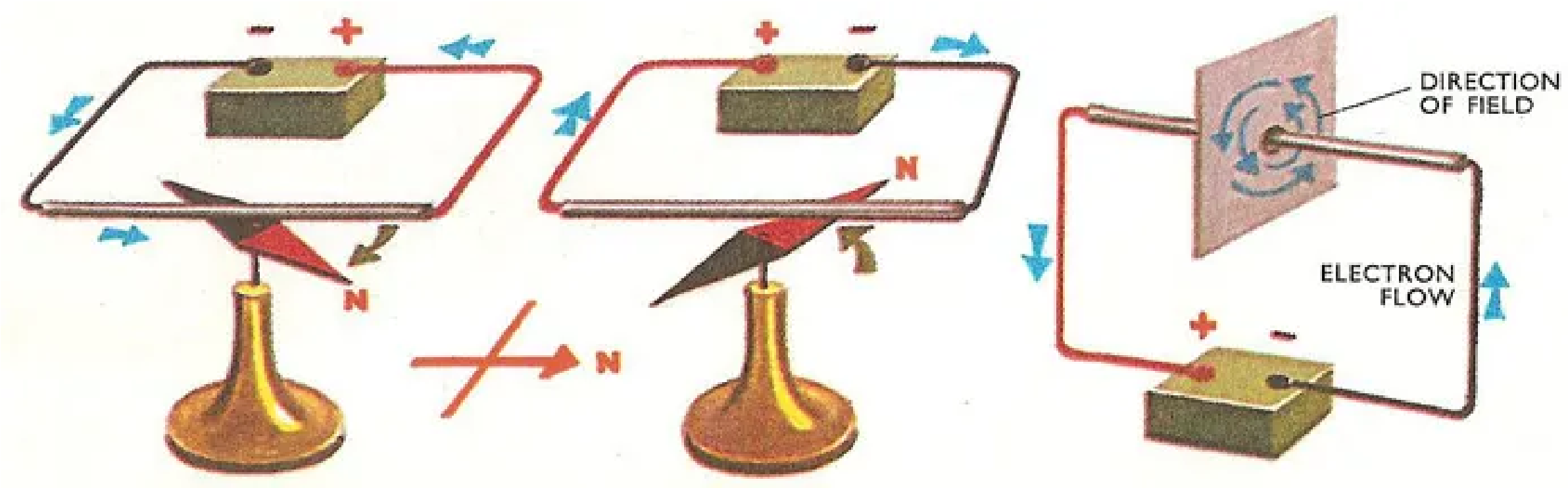
Close friend of a Hans Christian Andersen (1805 - 1875), author of many fairy tales, including "The Little Mermaid" and "The Ugly Duckling")

Hans Christian Ørsted (1777 - 1851)



First to explicitly describe and name the "thought experiment"

Hans Christian Ørsted (1777 - 1851)

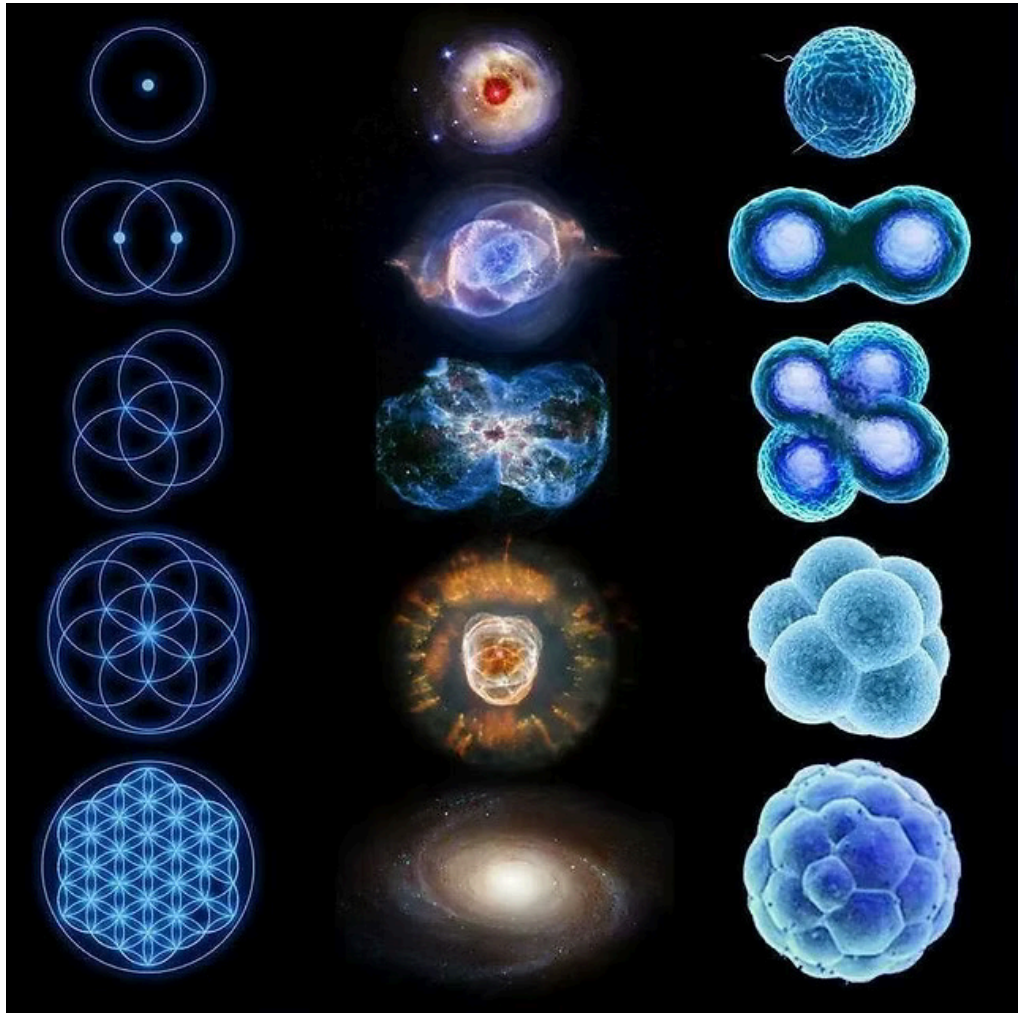


In 1820, Ørsted published his discovery that a compass needle was deflected from magnetic north by a nearby electric current, confirming a direct relationship between electricity and magnetism.

https://subratachak.wordpress.com/2017/12/16/oerstead/

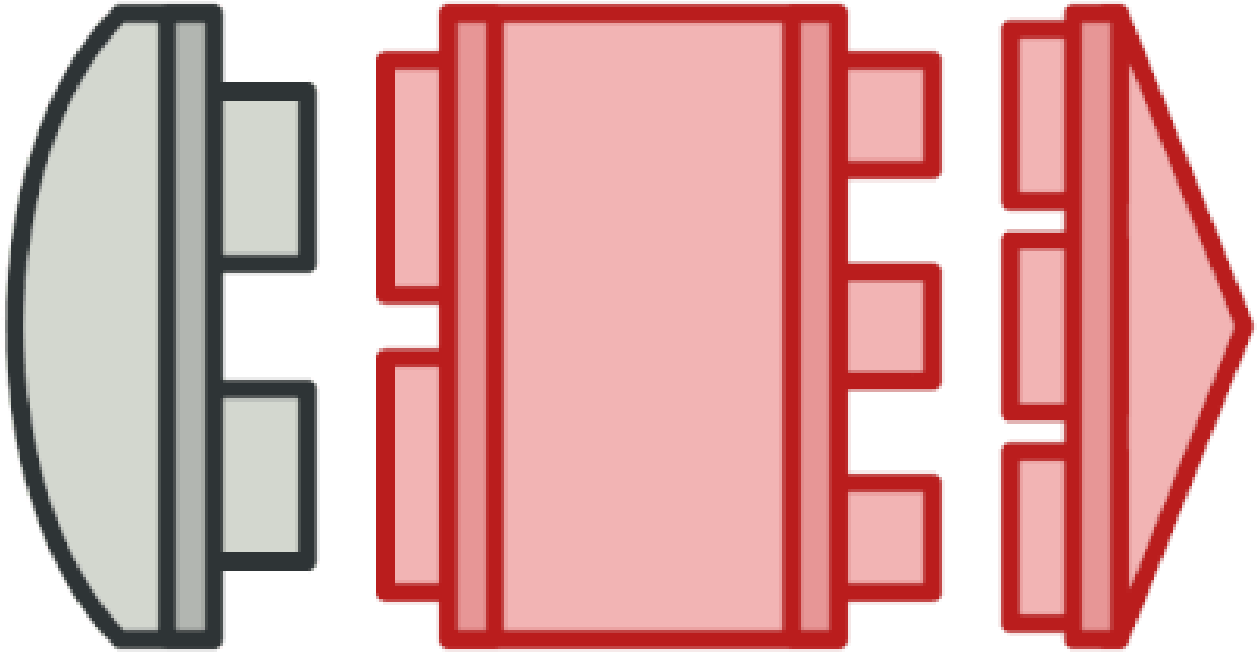
DefinIT philosophy

Definitions as "patterns in the universe"



Patterns in math, astronomy, biology.

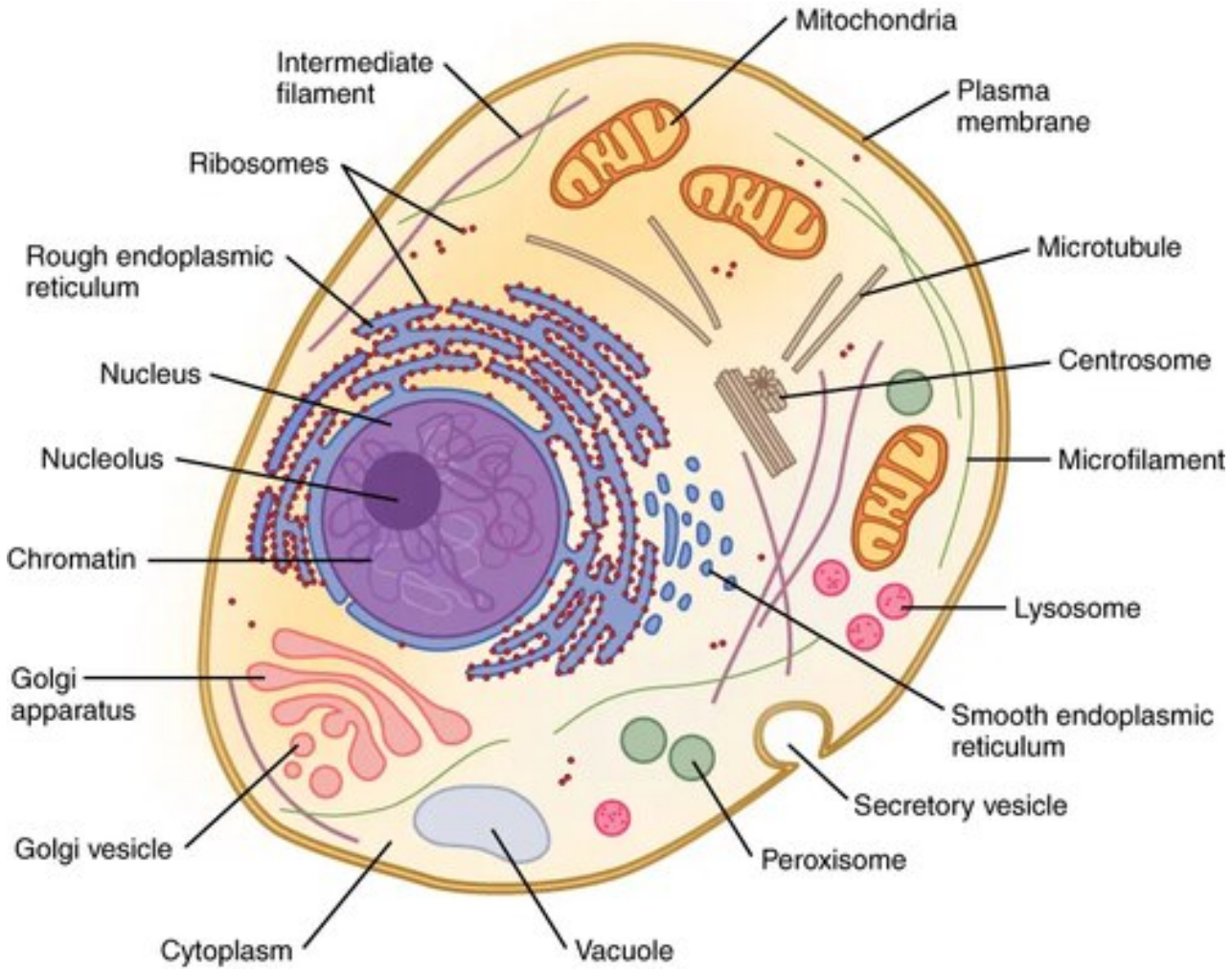
Definitions as "patterns in the universe"



A software design pattern.

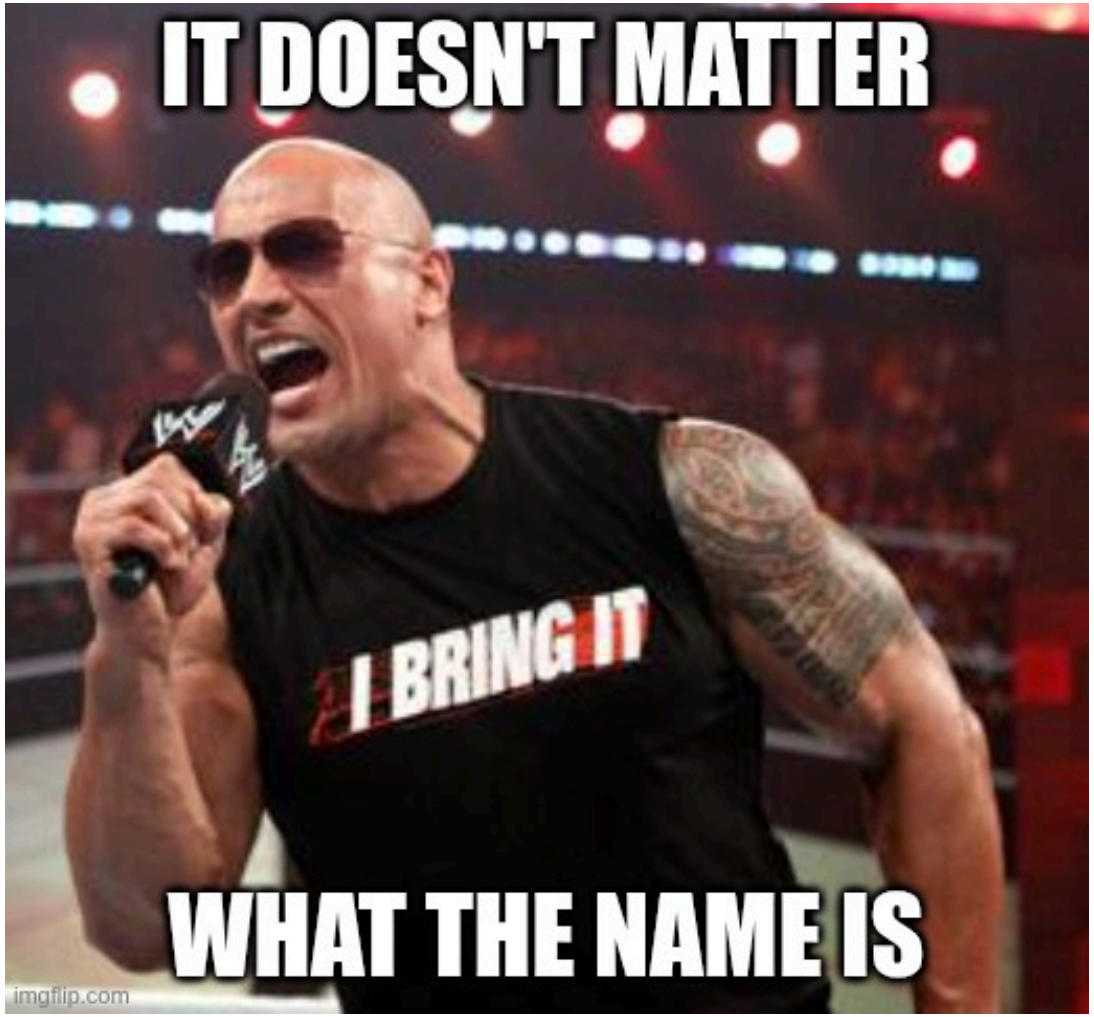
https://thecodinggopher.substack.com/p/the-adapter-design-pattern

The theory of acyclic references in terminology



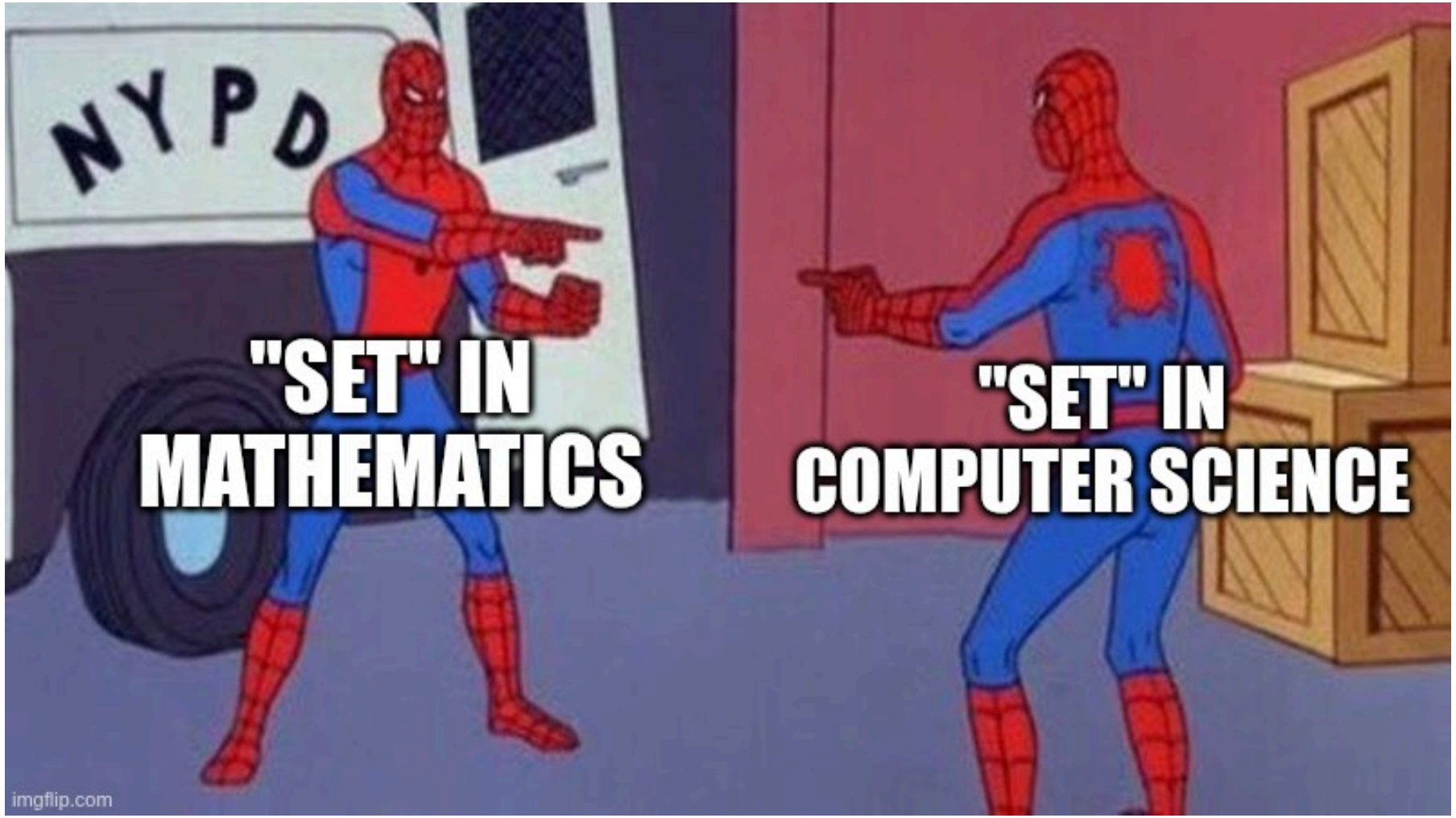
The exercise is to describe mitochondria without referring to the cell.

Naming is irrelevant



Naming does not matter, or at least is a secondary concern.

Languages are ambiguous



Even single words can have different meanings.

An experiment to "prove the theory"



"In theory, there is no difference between theory and practice. But in practice, there is." ~ Benjamin Brewster

Literature review

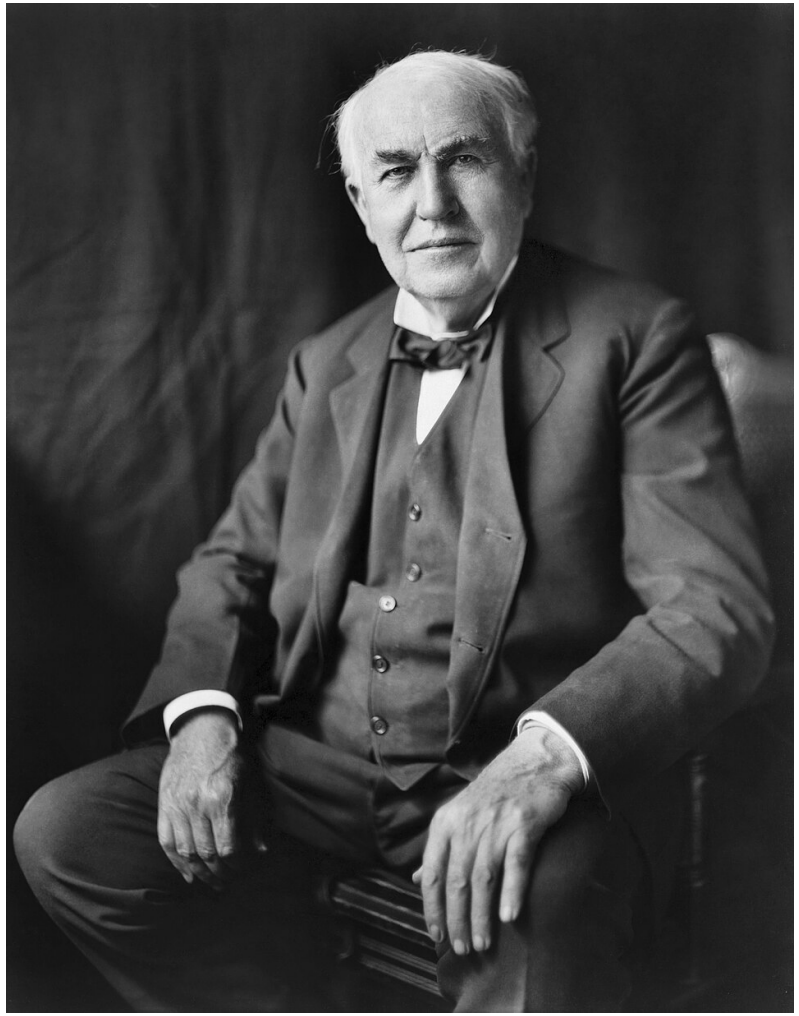
Standardized glossary in computing



A glossary is a focused collection of SPECIALIZED terms and DEFINITIONS tied to a SPECIFIC SUBJECT, book, or document.

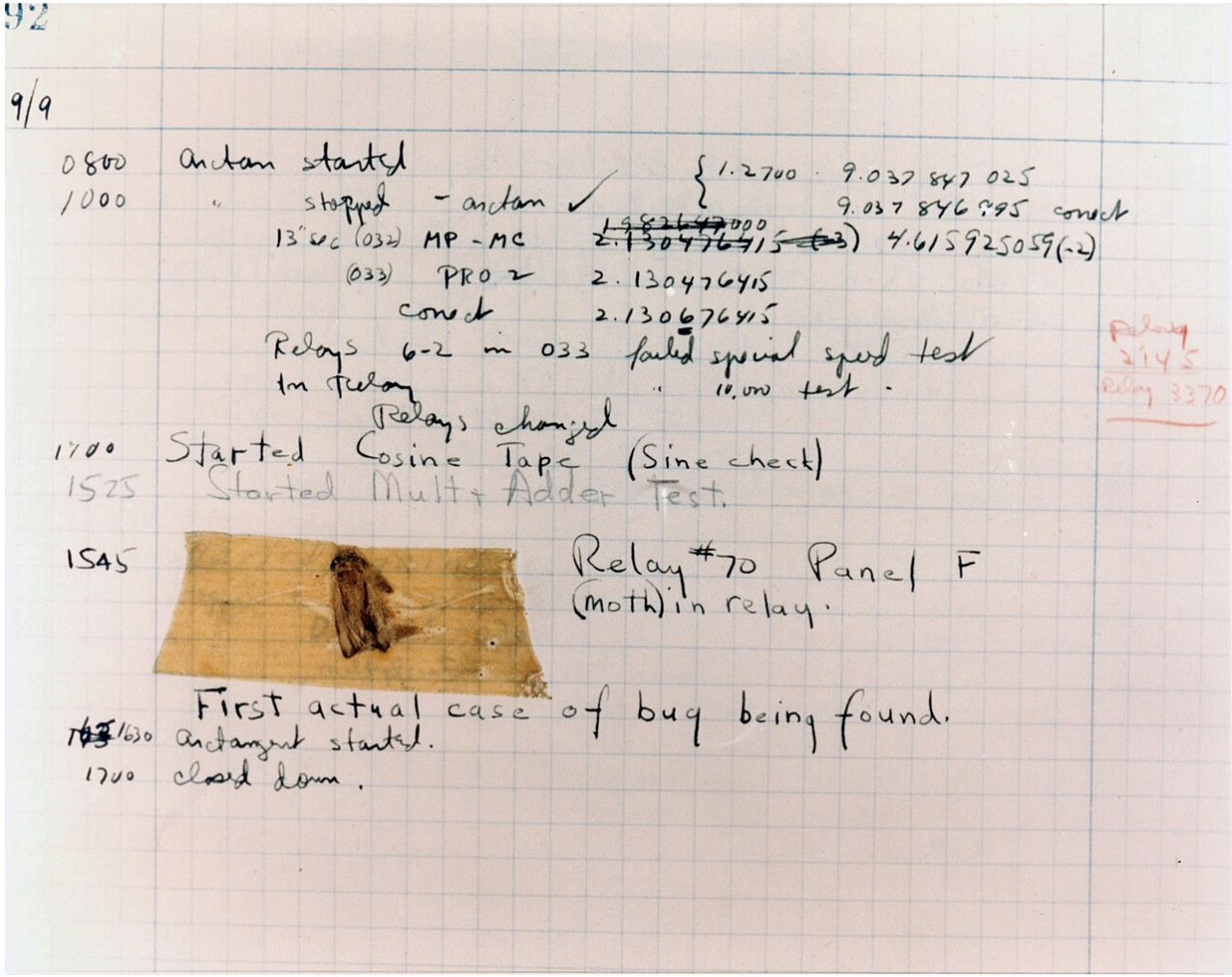
https://dataedo.com/cartoon/glossary-or-dictionary

Thomas Edison (1847 - 1931)



"Bugs" - little faults and difficulties

Grace Hopper (1906 - 1992)



"First actual case of bug being found"

Grace Hopper (1906 - 1992)



"First Glossary of Programming Terminology"

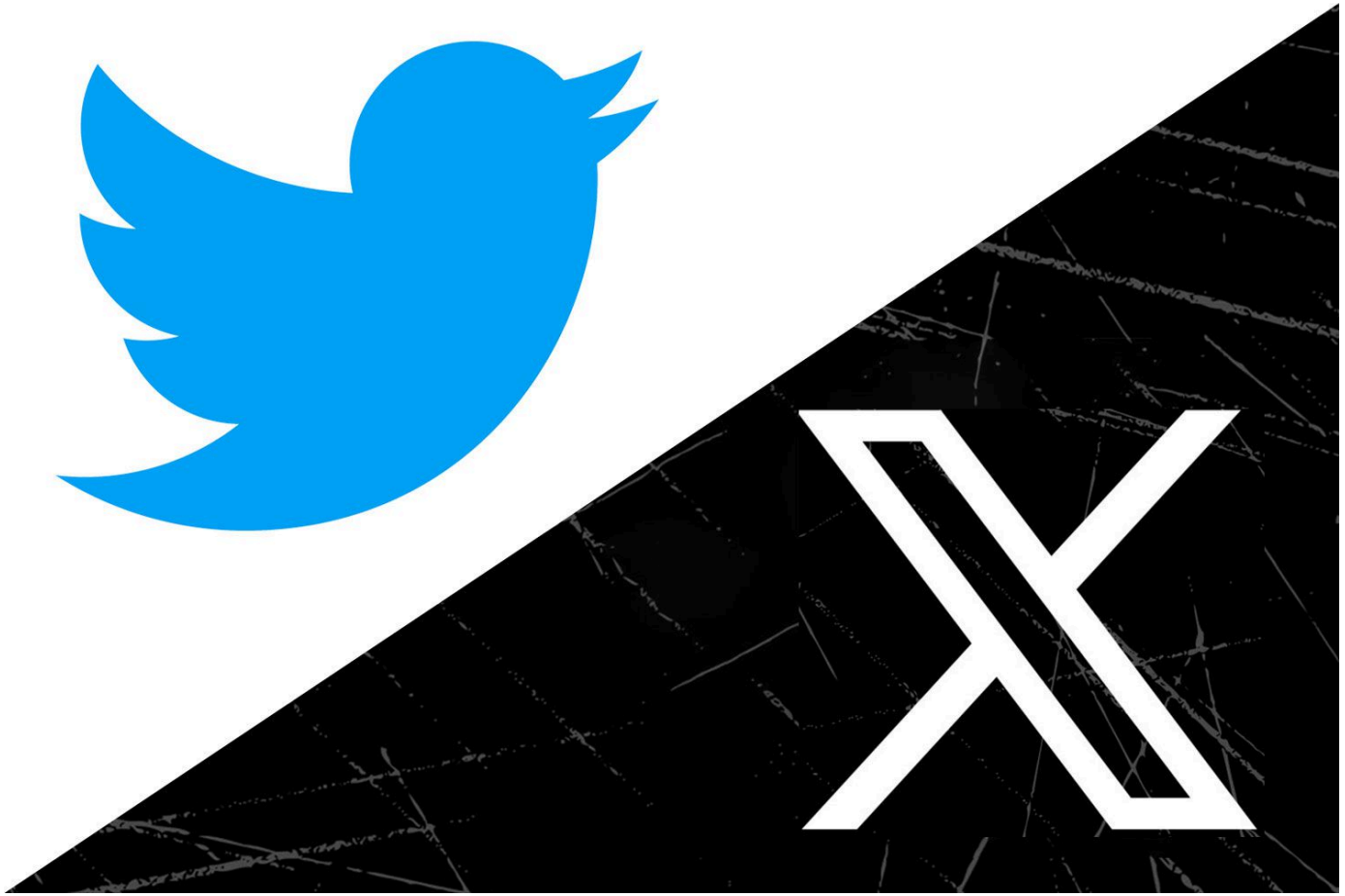
Eugen Wüster (1898 - 1977)





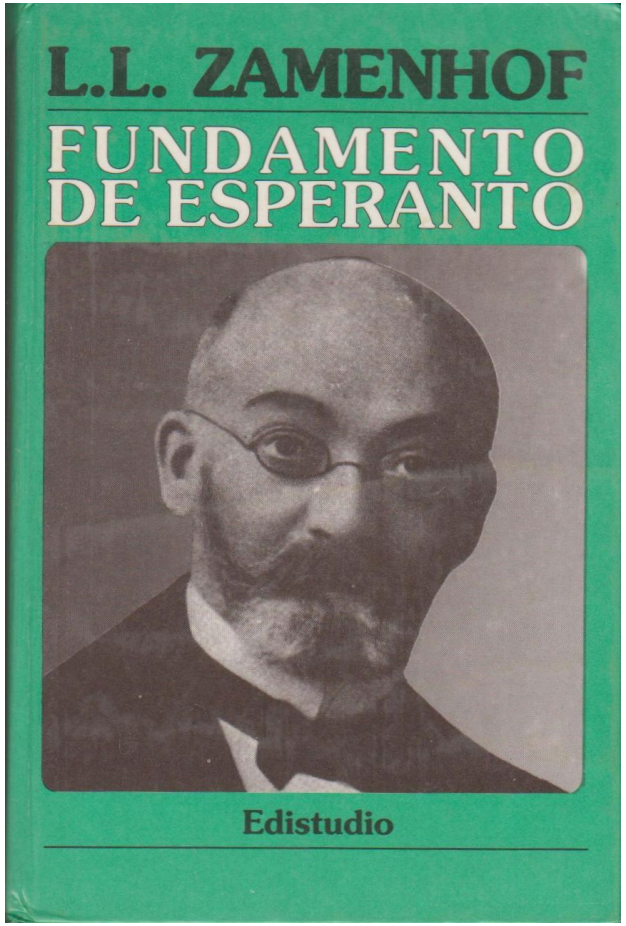
Eugen Wüster is often recognized as the founder of terminology.

Eugen Wüster (1898 - 1977)



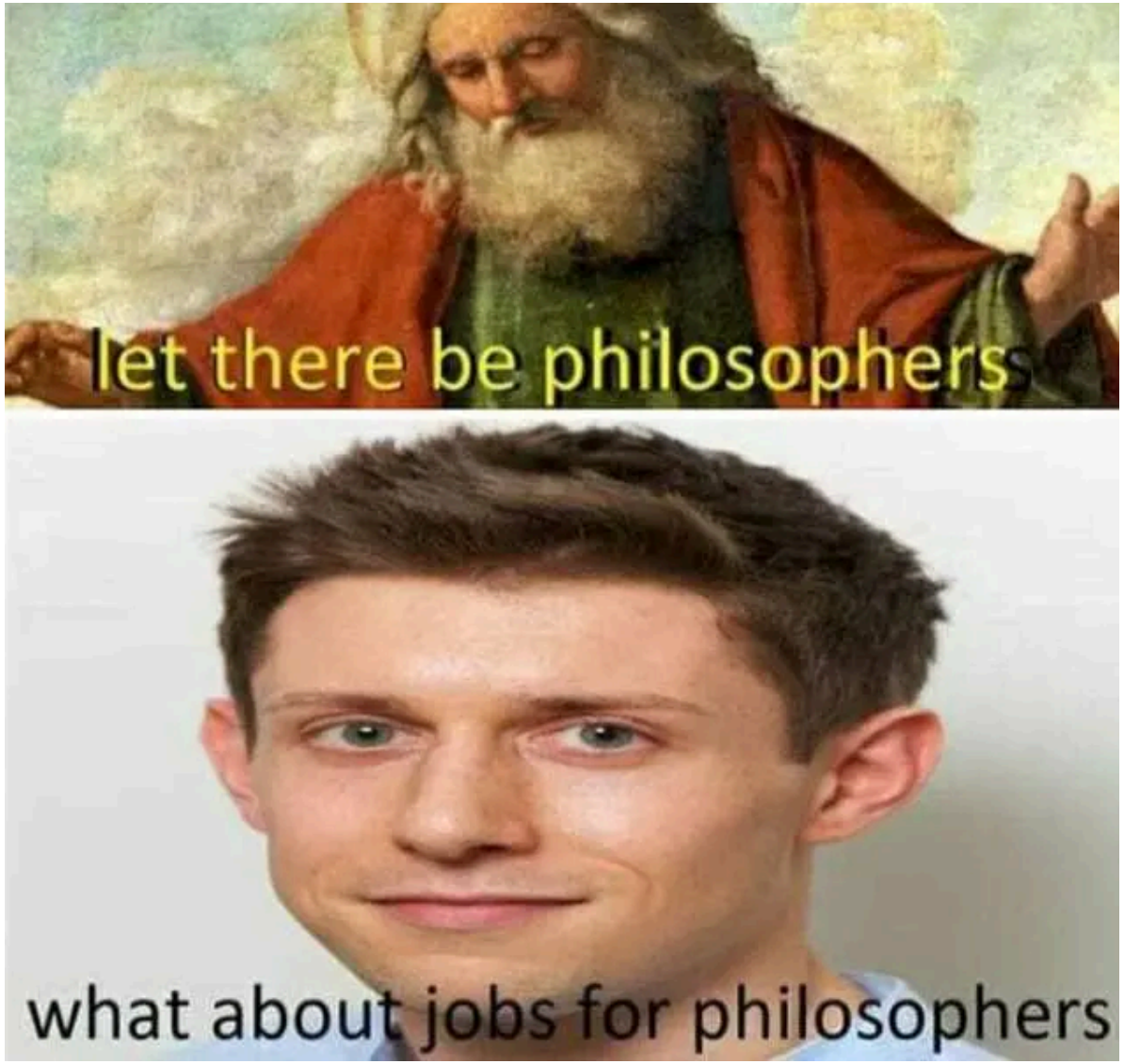
Concepts (or definitions) exist even before they are named - thus naming and it relevance for the terminology field have been neglected."

Ludwik Zamenhof (1859 - 1917)

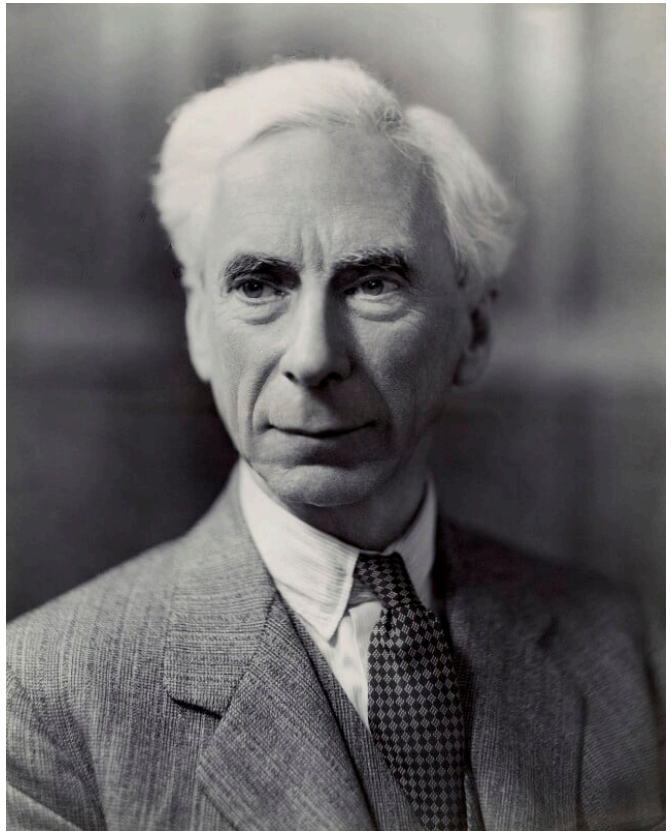


Wüster was a staunch advocate of a single language for scientific and technical communication.

Philosophy of definitions



Bertrand Russell (1872 - 1970)



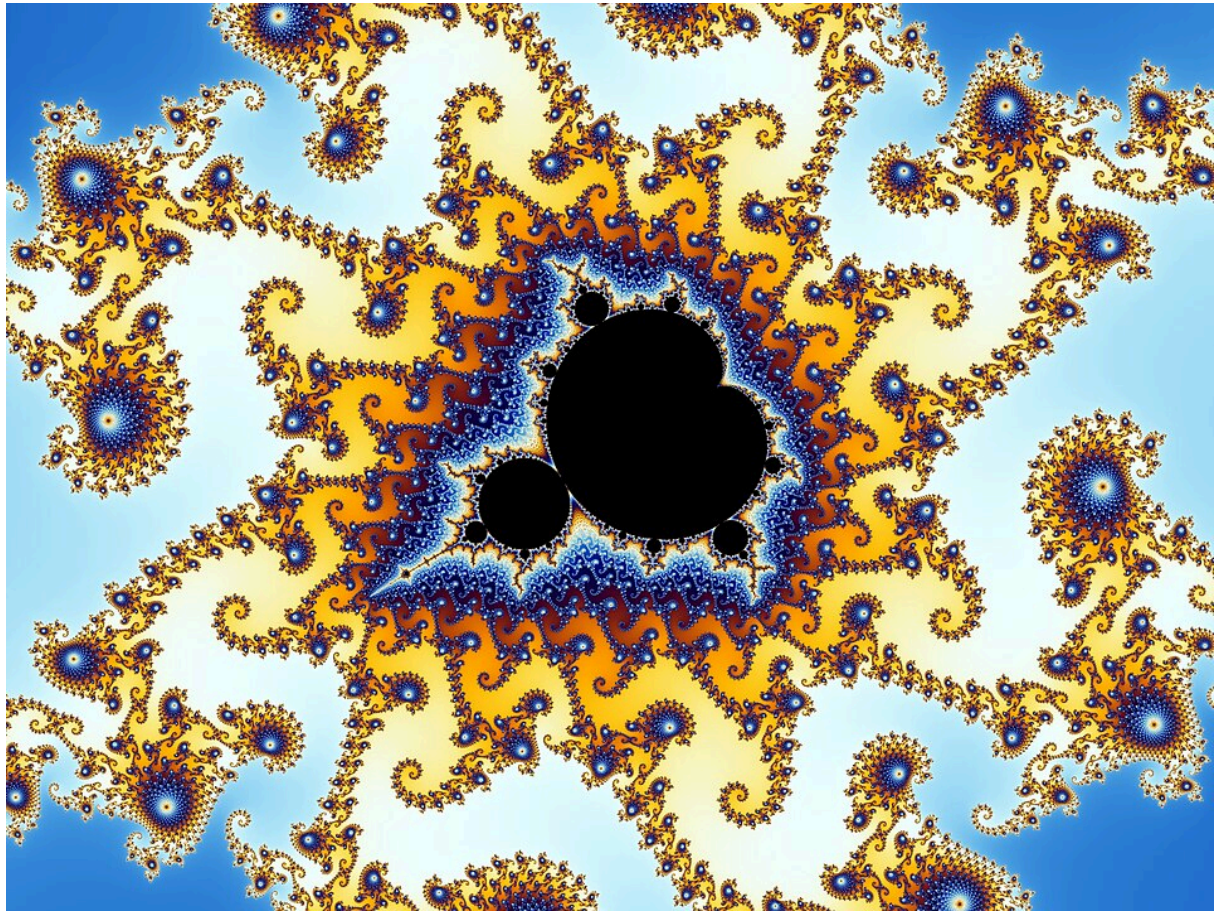
In 1950, Russell was awarded the Nobel Prize in Literature "in recognition of his varied and significant writings in which he champions humanitarian ideals and freedom of thought".

Logical atomism



The world is composed of independent, irreducible units of reality (logical atoms).

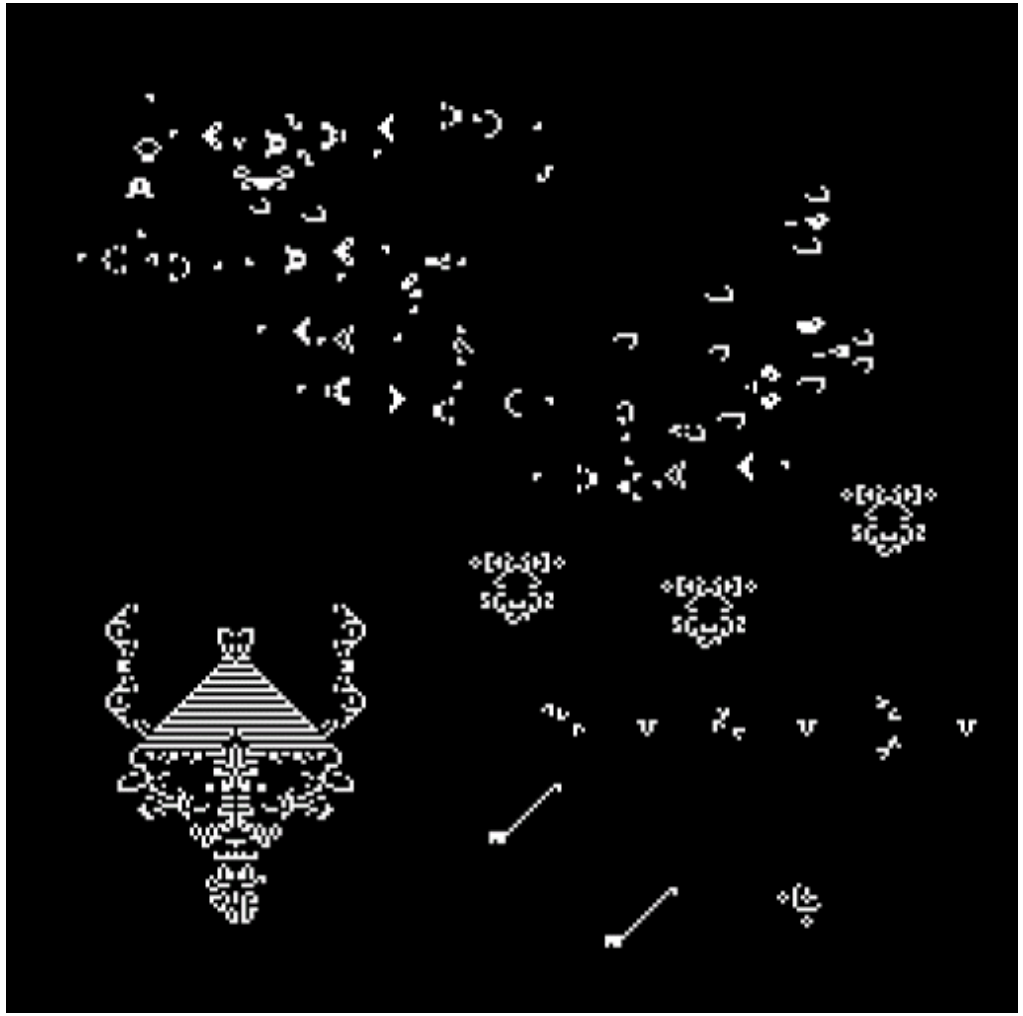
Benoit Mandelbrot (1924 - 2010)



Mandelbrot Set, being indisputably complex, uses a simple recursive equation:

$z_{n+1} = z_n^2 + c$, starting with $z_0 = 0$

John Conway (1937 - 2020)



Conway's Game of Life

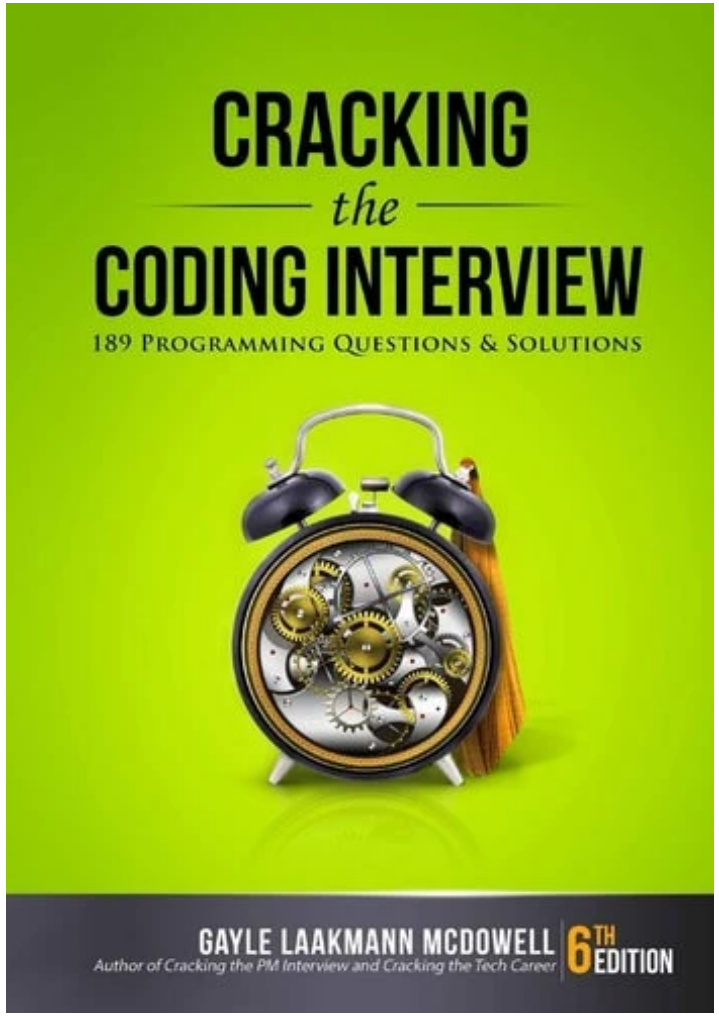
Mark Changizi (1969 - ?)

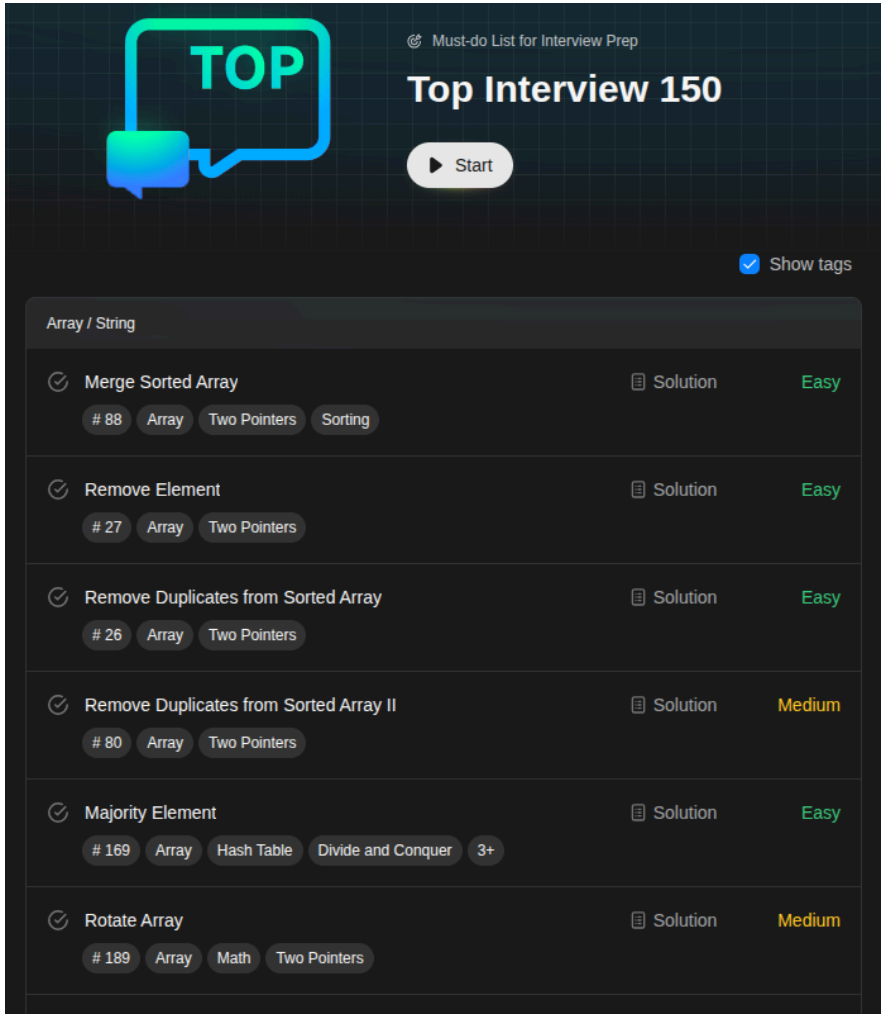
Organized hierarchies in Dictionaries

https://www.sciencedirect.com/science/article/abs/pii/S1389041708000120

Data Structures and Algorithms course

How it was created





Transform the content of a book or a course into DefinIT format.

DefinIT format

```python
from definit.definition.definition import Definition
from definit.definition.definition_key import DefinitionKey

from definit_db.data.field import FieldName
from definit_db.data.field.mathematics.definitions.fundamental.object import OBJECT
from definit_db.data.field.mathematics.definitions.fundamental.plane import PLANE
from definit_db.data.field.mathematics.definitions.fundamental.transformation import TRANSFORMATION


    Jan Bielecki, 2 months ago | author (Jan Bielecki)
class _Axis(Definition):
    def _get_content(self) -> str:
        return f"""
An axis is a reference line about which a geometric or coordinate space is organized. Positions may be measured
along an axis (as coordinates are along the x-axis and y-axis of a {PLANE.key.get_reference()}), and
{TRANSFORMATION.key.get_reference(phrase="transformations")} may be defined with respect to an axis (as a
reflection mirrors points across it). An axis is itself an
{OBJECT.key.get_reference(phrase="object")} of the space.

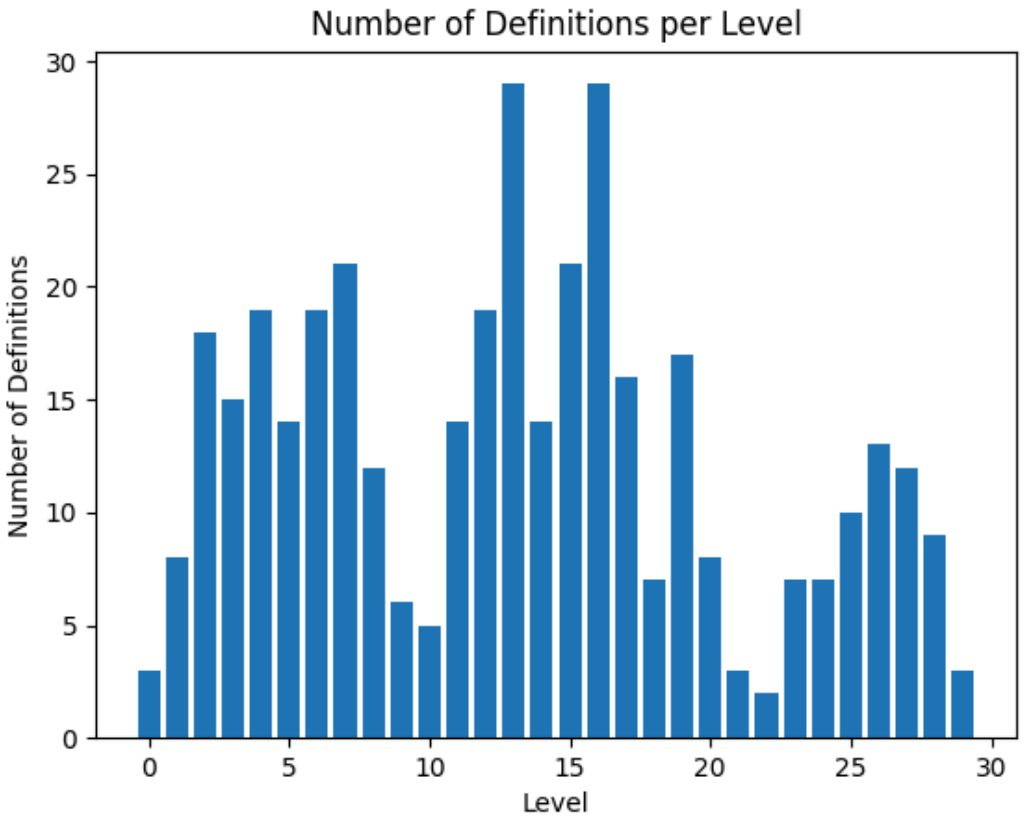
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In the two-dimensional plane, the x-axis and y-axis are two perpendicular reference lines. The point (3, 5) is
located three units along the x-axis and five units along the y-axis. Reflecting that point across the x-axis
gives (3, -5): the axis serves both as a basis for measuring position and as the line of reflection.
"""


AXIS = _Axis(
    key=DefinitionKey(
        name="axis",
        field=FieldName.MATHEMATICS,
    )
)
```

Definition as a Python module.

DSA course statistics



Word	Occurrences	Distinct definitions
every	100	70
left	61	29
exactly	51	38
given	50	46
position	50	27
result	47	36
represent	46	33
since	45	39
example	44	42
produce	44	36

30 levels, 380 definitions, ~1700 unique non-definition words and 1783 references.

Issues

Jan 2026	Gemini 3 Pro	GPT-5.2	Claude Opus 4.5	Kimi K2.5	Grok-4
Feb 2026	Claude Opus 4.6	Gemini 3.1 Pro	Grok 4.20	Gemini 3 Pro	GLM-5
Mar 2026	Claude Opus 4.6 (Thinking)	Gemini 3.1 Pro	GPT-5.4	Grok 4.20	MiniMax M2.7
Apr 2026	Claude Opus 4.6 (Thinking)	Claude Opus 4.7	Meta Muse Spark	Gemini 3.1 Pro	GPT-5.5
May 2026	Claude Opus 4.6 (Thinking)	Claude Opus 4.7	Meta Muse Spark	Gemini 3.1 Pro	GLM-5.1 / Qwen3.7-Max
Jun 2026	Claude Opus 4.6 (Thinking)	Claude Fable 5	Gemini 3.1 Pro	GPT-5.5	GLM-5.2 / DeepSeek V4.1 Pro
Jul 2026	Claude Opus 5 (Max)	Claude Fable 5	Claude Opus 4.6	GPT-5.6 Sol	Kimi K3 / Grok 4.5
Aug 2026	Claude Opus 5 (Max)	Claude Fable 5	Claude Opus 4.6	Meta Muse Spark 1.2	Qwen3.8-Max / DeepSeek V4 Pro
Sep 2026	Gemini 4 Argon (High)	Claude Opus 5.5	Claude Fable 5.1	Claude Opus 5	GPT-6 Astra
Oct 2026	Gemini 4 Carbon (High)	Gemini 4 Argon (High)	Claude Opus 5.5 (High)	Meta Muse Wonder (Max)	Claude Fable 5.1 (Max)

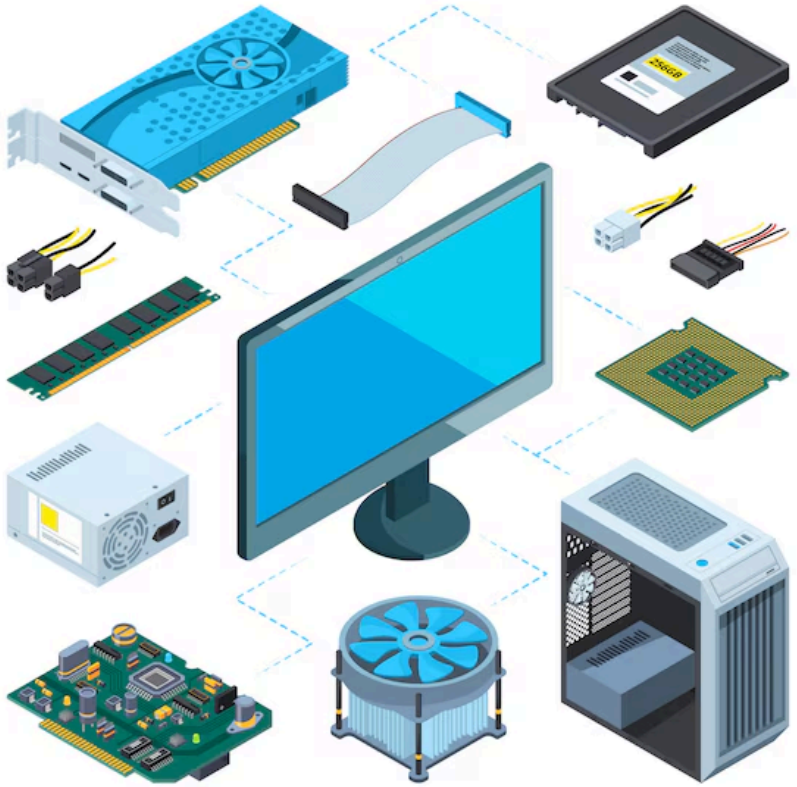
Not proven, not used, not easy.

Further work

Write a paper



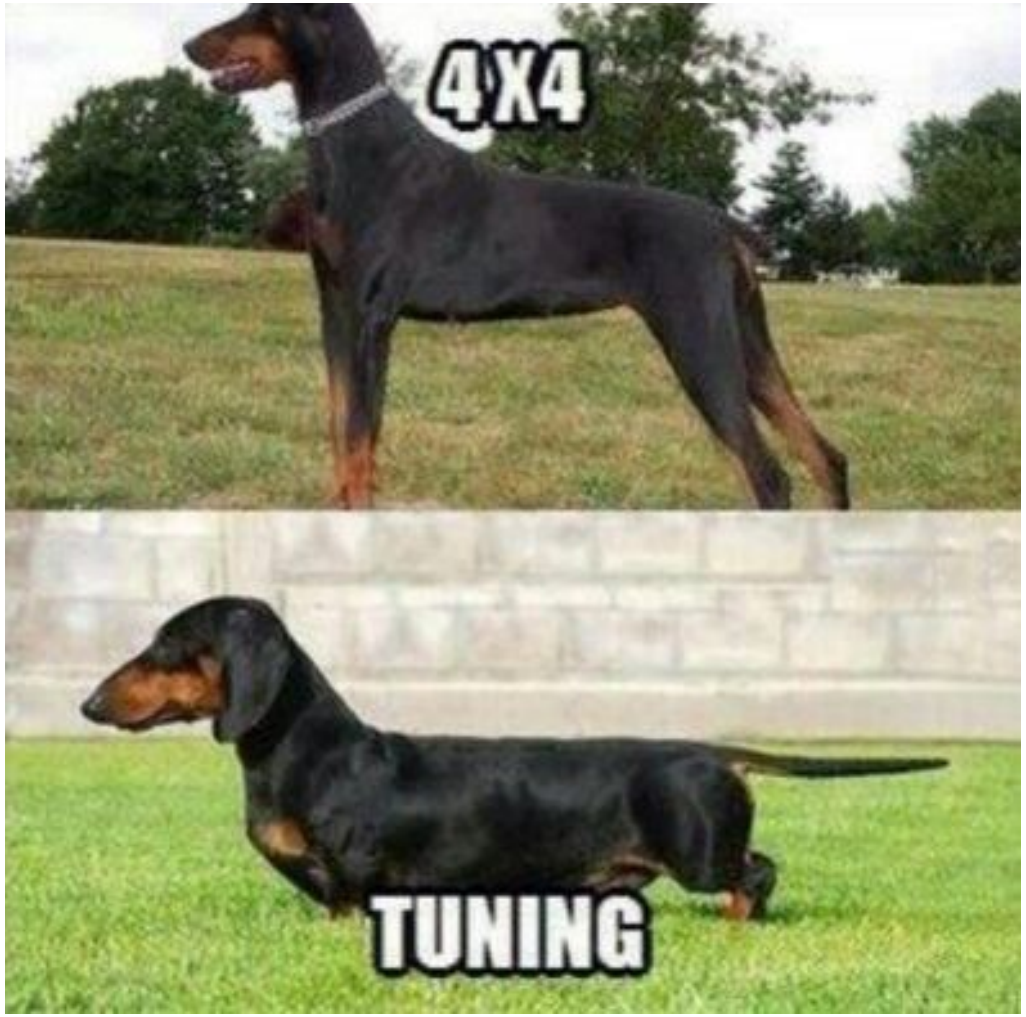
Create the next course





Mind the "abstraction level" of the course, it should be just above the level of DSA course, so it can reuse the same definitions.

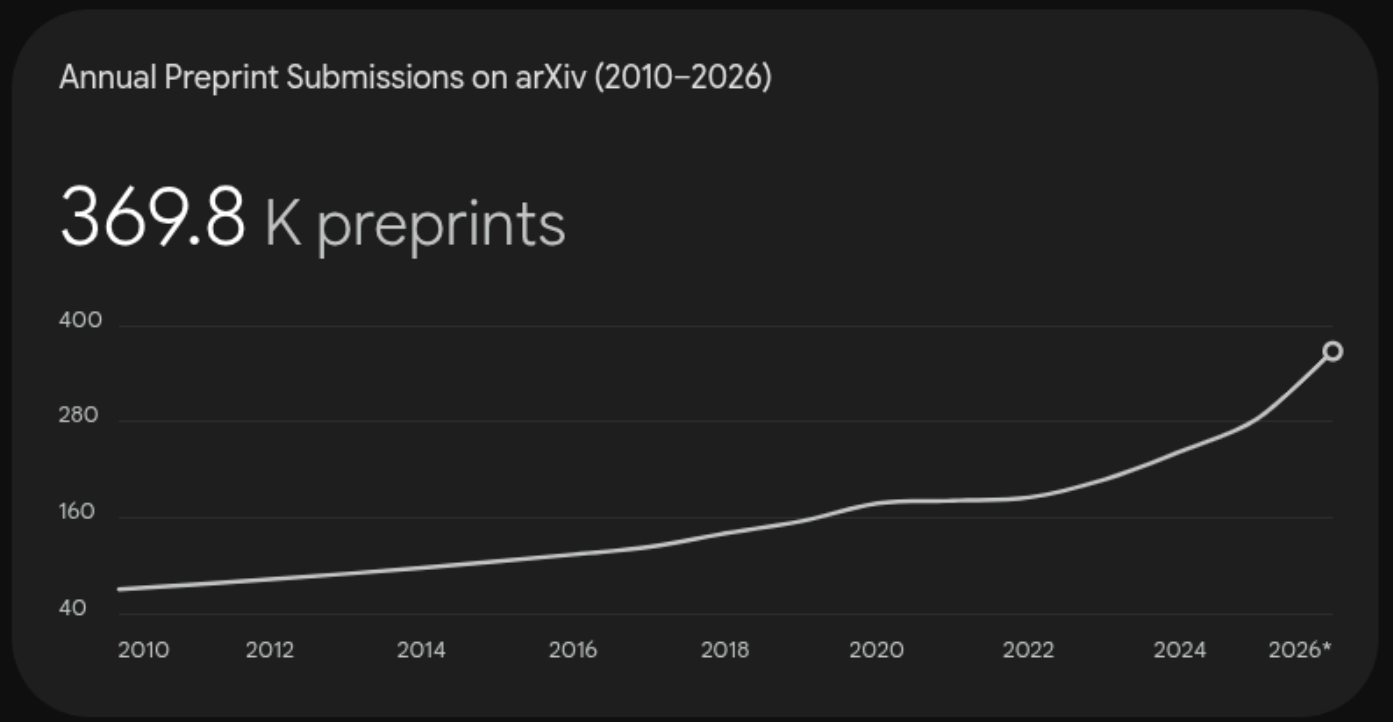
Other applications



Data from LLMs, for LLMs. Fine-tuning, RAGs.

Other applications

Focusing on the 2010–2026 window highlights the post-2022 LLM inflection point even more clearly. Between 2010 and 2020, arXiv preprint submissions grew steadily from 70,100 to 178,300 per year before plateauing during 2021–2022 at just ~2% annual growth. Beginning in 2023, annual volume broke sharply upward—more than doubling the pre-LLM plateau pace to reach an estimated **369,800 submissions in 2026.**



"New kind of science" -> reducing the number of preprints, articles, books.

Please generate a thank you slide for the audience. Don't make a mistake.